Tianhao Fei

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Personal Information

○ **Date of Birth:** March 6, 2002 ○ **Marital Status:** Single ○ **Gender:** Male ○ **Home:** Sichuan, China

Education

Master, University of Science and Technology Beijing *September 2024 – June 2027*

School of Automation and Electrical Engineering, Control Engineering

- **Score:** 90.1/100; **GPA:** 3.9/4.0; **Rank:** 3/175 (top 2%); **Supervisor:** **Yongliang Yang**.
- **Selected Courses:** Global Perspectives(94); Motor Control and Design(93); English(92); Engineering Optimization Methods(92) System Stability Analysis(92); Active Disturbance Rejection Control(92); Patter Recognition(91); Numerical Methods(90).

Bachelor, University of Science and Technology Beijing *September 2020 – June 2024*

School of Automation and Electrical Engineering, Automation

- **Score:** 80.9/100; **GPA:** 3.22/4.0.

Research Interests

Unmanned Systems, Embodied Intelligence, Robotics, Reinforcement Learning, Adaptive Dynamic Programming, Optimal Control.

Research Experience

1. **Adaptive Backstepping Control for Constrained Uncertain Nonlinear Systems**

- **Adaptive control under multiple uncertainties:** Developed backstepping-based neural control schemes for nonlinear contrained systems with unknown parameters, stochastic disturbances, and unknown control directions.
- **Distributed synchronization and robust tracking:** Designed command-filtered and compensation-based controllers for heterogeneous switched multi-agent systems and time-delay dynamics under finite-energy disturbances.
- **Stability and performance guarantees:** Established boundedness, constraint satisfaction, bipartite synchronization, and robust H_∞ tracking performance through Lyapunov-type, and zero-sum game formulations.
- **Outcome: 3 journal articles** (*Neurocomputing*, *Journal of the Franklin Institute*, *International Journal of Robust and Nonlinear Control*, Published).

2. **Projected Self-Triggered Reinforcement Learning for Differential Games**

- **Self-triggered learning:** Proposed communication-efficient learning frameworks for zero-sum and nonzero-sum differential games, avoiding continuous state monitoring while preserving closed-loop stability.
- **Projected safe adaptation:** Incorporated projection operators into Actor–Critic and Actor–Critic–Disturbance learning laws to prevent parameter drift and keep adaptive weights within prescribed compact sets.
- **Resource-aware Nash approximation:** Combined intermittent feedback, fuzzy logic, and experience replay to approximate Nash policies with reduced sampling, bounded learning errors, and Zeno-free execution.
- **Outcome: 2 journal articles** (*IEEE Transactions on Emerging Topics in Computational Intelligence*, *IEEE Transactions on Fuzzy Systems*, Early Access); **3 conference papers**.

3. **Data-Efficient Composite Actor–Critic Reinforcement Learning**

- **Composite learning for game-theoretic problem:** Developed Actor–Critic learning algorithm for multi-player nonzero-sum differential games by integrating instantaneous Bellman errors with historical data.
- **Relaxed excitation requirements:** Replaced restrictive persistence of excitation condition with finite/sufficient excitation-based learning, improving the practicality of online adaptive dynamic programming.
- **Improved convergence and stability:** Established convergence of critic/actor weights and closed-loop stability while enabling approximate Nash equilibrium learning for nonlinear multi-agent decision systems.
- **Outcome: 1 journal article** (*IEEE Transactions on Neural Networks and Learning Systems*, Under Review).

Publications

Journal:

1. **T. Fei**, *et. al*, "Game-Theoretic Fuzzy Reinforcement Learning via Self-Triggered Mechanism". IEEE Transactions on Fuzzy Systems, early access, 2026, doi: 10.1109/TFUZZ.2026.3706062. **(JCR Q1)**
2. **T. Fei**, *et. al*, "Self-Triggered H_∞ Control Using Projected Actor-Critic-Disturbance Learning". IEEE Transactions on Emerging Topics In Computational Intelligence, early access, 2026, doi: 10.1109/TETCI.2026.3683743. **(JCR Q1)**
3. **T. Fei**, *et. al*, "Adaptive Neural Design for Permanent Magnet Synchronous Motor with Asymmetric Constraints and Input Dead-Zone," Neurocomputing, vol. 640, p. 130294. **(JCR Q1)**
4. **T. Fei**, *et. al*, "Command Filtered Adaptive Bipartite Synchronization for Heterogeneous Switched Multi-Agent Systems with State Delays and Input Nonlinearity," Journal of the Franklin Institute, vol. 363, no. 2, p. 108395. **(JCR Q1)**
5. **T. Fei**, *et. al*, "Robust Adaptive H_∞ Tracking Control for Nonlinear Strict-Feedback System via Actor-Critic-Disturbance Learning: A Zero-Sum Game-Based Approach," International Journal of Robust Nonlinear Control, vol. 36, no. 8 , pp. 4517-4536. **(JCR Q1)**

Conference:

7. **T. Fei**, *et. al*, "Self-triggered Sampling Design For Continuous-Time Linear Dynamical System," 2025 China Automation Congress (CAC), Harbin, China, 2025, pp. 3164-3169.
8. **T. Fei**, Y. Yang*, "Self-Triggered Nash Equilibrium Seeking for Multi-Player Linear Quadratic Differential Games," 2026 Chinese Control and Decision Conference (CCDC), Nanjing, China, 2026, pp. 3445-3450. **Accepted, Oral presentation**.
9. **T. Fei**, *et. al*, "Communication-Efficient Reinforcement Learning for Linear Quadratic Differential Zero-Sum Games, " 2026 International Annual Conference on Complex Systems and Intelligent Science (CSIS-IAC), Hefei, China, 2026, pp. 491-496. **Accepted, Best Paper Award**.

Patent:

10. Y. Yang, **T. Fei**, *et. al*, "Self-Triggered Robust Reinforcement Learning Method and Apparatus for Zero-Sum Games of Unmanned Surface Vehicles". Publication No. CN122151484A.
11. Y. Yang, **T. Fei**, *et. al*. "Resilient Control for Power Buffers in DC Microgrids Against Denial-of-Service Attacks". Publication No. CN121566407A.
12. Y. Yang, Z. Shan, **T. Fei**, *et al*. "Composite Game-Theoretic Reinforcement Learning for Quadrotor UAV Swarms". Publication No. CN121879395B.
13. Y. Yang, Z. Shan, **T. Fei**, Q. Li. "Secure Control Method and Device for Power Buffers in DC Microgrids Against FDI Attacks". Publication No. CN121440520A.

Honors and Awards

- **2026 First Prize Scholarship for Master Students**, USTB. **(TOP 2%)**
- **2026 "Pillars of the Nation" Elite Talent Award**, China Youth Development Foundation. **(150 recipients nationwide)**
- **2026 Best Paper Award**, International Annual Conference on Complex Systems and Intelligent Science.
- **2026 Commemorative Certificate for Research Achievement in Invention Patent**, "Guohong Zhongjing Cup", School of Automation and Electrical Engineering, University of Science and Technology Beijing (USTB).
- **2025 Excellent Merit Master Student**, USTB. **(TOP 5%)**
- **2025 Inspur "Langchao" Enterprise Scholarship**, Inspur Group Co., Ltd. **(16 recipients university-wide)**
- **2025 First Prize Scholarship for Master Students**, USTB. **(TOP 7%)**
- **2023 Second Prize, Mathematical Competition** of USTB.
- **2022 Social Practice Gold Award**, USTB.

Skills

Languages Mandarin Chinese (Native), English (CET-6, 529).
Coding Matlab, Python, LaTeX…